

high-efficiency particulate air (HEPA) filters are the preferred choice due to their advanced purifying process. The popularity of smart air-purifiers is also growing as they can be conveniently controlled via Wi-Fi and Bluetooth remotely.

In developing countries like India, despite high levels of pollution, the purifier market is still in its nascent stage due to high initial cost and addi-

tional cost of replacing filters every few months. Also, the alternative charging efficiency of solar purifiers is not that high. However, their sales are expected to surpass those of developed countries in the coming years, according to a recent report from Research and Markets.

Using solar power is not the only sustainable way when it comes to air-purifiers. Recently, student

researchers from Lovely Professional University and Indian Institutes of Science Education and Research (IISER), Mohali, have developed an algae-based air-purifier that increases the amount of oxygen through photosynthesis besides neutralising the air pollutants. It also produces biomass as byproduct, which can be used for energy generation.

—Ayushee Sharma

3 AI-POWERED CAMERAS FOR SECURITY

AI-enabled security cameras can send alerts when an object is missing from its usual place, or a new unknown object has been lying around for too long. Monitoring officers can check whether the activity is a threat or not

Increasingly complex crime scenarios and other security threats have led to the widespread use of CCTV cameras and video surveillance systems in commercial, residential and all other places that need monitoring. But for mission-critical applications that require reliable scrutiny constantly, these traditional security cameras do not suffice.

Main issue with these cameras is that limited object detection puts the onus on human operators to be attentive to the recording for preventing intrusion and other unwanted incidents. Otherwise, the observations can only be made after the mishap has already occurred. This necessity of automation has given rise to cameras and surveillance systems that utilise artificial intelligence (AI) and machine learning to provide real-time alerts, even in harsh weather conditions, in case of actual emergencies. In most models, features like voice and facial recognition, integration with assistants like Alexa, gesture control are also enabled.

Based on the scenario, rule-based AI systems or behavioural learning approach may be required. In rule-based AI, predefined instructions are set by the programmer for such



(Credit: www.scienceabc.com)

purposes as defining restricted areas and varying what should be considered unsuspicious for a particular time of the day. Alerts in the form of notifications and video clips are sent if the rules are violated. Whereas for complex or busy environments, behavioural analytics is preferred as it uses self-learning based on its perception of environment.

Broad categorisation of objects is done based on characteristics like size, shape, colour, motion, angle, and matches with deep learning algorithms initially, which narrows down with usage over time to precisely differentiate unusual from normal. For example, alerts can be sent when an object is missing from its usual place or a new unknown object has been lying around for too long. Monitoring officers can check whether the activ-

ity is a threat or not. There are numerous companies like IntelliVision, Clarifai and Valossa that provide AI-based video analytics and software for smart cameras.

These abilities make AI-based cameras particularly useful for safeguarding national and international interests by aiding the security forces in making smart and strategic decisions in times of threats like armed aggression and terrorism. For instance, the

recently launched AI camera SIP-2HDG-W40 by Secureye has been deployed by national security forces in a region in North India which can start recording automatically when required.

It comes with a 2MP Sony IMX307 Starvis CMOS sensor and has an infrared range of thirty metres. Salient features include intelligent motion detection (like tripwire to detect entry and exit or heat mapping to identify hotspots), two-way communication via intercom, support for both 4G and Wi-Fi connectivity, inbuilt as well as cloud storage, IP66 waterproof rating. Secureye Falcon app available on both Android and iOS enables remote monitoring. With smart dual light source, it can record HD (1920 x 1080 pixels) or SD (640 x 360 pixels) footage with a frame rate of 15fps, even in